Advanced statistics

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Hypothesis Testing

- Its a statistical method, used to evaluate assumptions about population based on sample data.

- It estimates whether the change in population data has occurred because of any decision or not.

Steps in HT

1) Formulating the hypotheses

- we need to decide <u>Null Hypothesis H_0</u> and <u>Alternate Hypothesis H_1</u>

H_0 - represents current state or no effect state.

H_1 - represents the state where different or effect is observed.

eg. We implemented new teaching method in school to improve their board marks.

→ current avg marks = 70
after new method avg marks = 75

$H_0 : \mu \leq 70$ $H_1 : \mu > 70$

- For a given HT, find H_1 using state of difference or effect and then use its opposite condition as H_0

eg.

State of effect	State of no effect
$H_1 : \mu > x$	$H_0 : \mu \leq x$
$H_1 : \mu < x$	$H_0 : \mu \geq x$
$H_1 : \mu \neq x$	$H_0 : \mu = x$

Step 2 : Collect sample for state of effect.

- Take a random sample of size n from population
- compute sample mean $\bar{x}$ as observed statistic

Step 3 : Compute test statistics

- The test statistic measures how far sample statistic (state of effect) deviates from current state or Null hypothesis H_0.

1) Z statistics : Z score

used when : $n \geq 30$

σ ie population std dev is known

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

if σ is unknown but $n \geq 30$ then

$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$s \approx \sigma$ when sample size is $\underline{\underline{n \geq 30}}$

2) T statistic : Tscore

used when: $n < 30$

: population std dev is unknown

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \times \text{factor}$$

decided based on sample size.

Step 4: Choosing the Significance Level (α)

- α represents a threshold for rejecting H_0

If the effect of decision on population is significant enough then only we can say the decision is effect and H_1 (state of effect) is True

otherwise if H_1 is <u>not true</u> means H_0 is <u>true</u>

if H_0 is true → it means no effect on population after decision.

- popular values of α are 0.05 / 0.1

5% 10%

confidence level = 95% 90%

step 5: Applying the decision rule.

$H_1: \mu > \bar{x}$ $H_0: \mu \leq \bar{x}$ - right tailed test

$H_1: \mu < \bar{x}$ $H_0: \mu \geq \bar{x}$ - left tailed test

$H_1: \mu \neq \bar{x}$ $H_0: \mu = \bar{x}$ - two tailed test

we have already computed Z-statistic / t-statistic for sample. and we've also specified the significance level α

In case of Z statistic

1. Right tailed test

$$Z > Z_\alpha \Rightarrow \text{Accept } H_1 \text{ or Reject } H_0$$

eg. New teaching method.

older avg marks = 70

sample avg marks after new method = 75

$n = 50, \quad \bar{x} = 75, \quad s = 8$

$H_1: \mu > 70$ $H_0: \mu \leq 70$ right tailed test

$\alpha = 0.05$

$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{75 - 70}{\frac{8}{\sqrt{50}}} = Z$$

get Z score wrt $\alpha = 0.05$

$\downarrow$

Z_α

compare Z and Z_α

if $Z > Z_\alpha \Rightarrow$ Accept H_1 or reject H_0

$Z \leq Z_\alpha \Rightarrow$ Do not Accept H_1 or Failed to reject H_0

2 : left tailed test

$Z < -Z_\alpha \Rightarrow$ Accept H_1 or reject H_0

eg. A new odd even vehicle formula is applied and polution has reduced.
We need to check if this change is significant enough or not.

$\longrightarrow$ prev. polution level : $\mu = 300$

$H_1 : \mu < \bar{x}$ $\qquad$ $H_0 : \mu \geq \bar{x}$ $\qquad$ left tailed test

collect sample : $n = 50$ $\quad \bar{x} = 280$ $\quad s = 10$

$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

set significance level to 0.1 and compute Z_α

compare Z and Z_α

if

$Z < -Z_\alpha \Rightarrow$ Accept H_1 or reject H_0

- the new formula has reduced polution significantly

if $Z \geq -Z_\alpha \Rightarrow$ Do not accept H_1 or Failed to reject H_0

$\rightarrow$ reduction in polution due to new method is not significant.

3: Two tailed test

$|Z| > Z_{\frac{\alpha}{2}} \Rightarrow$ Accept H_1 or Reject H_0

eg. We tested car engine with new ethanol mixed fuel. We want to check if it caused any change in engine mileage.

prev mileage : $\mu = 20$

sample $\rightarrow$ $n = 50$, $\bar{x} = 21.5$, $S = 2$

significance level = 2% = 0.02

$H_1 : \mu \neq \bar{x}$ $\qquad$ $H_0 : \mu = \bar{x}$

compute Z and $Z_{\frac{\alpha}{2}}$

$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{21.5 - 20}{\frac{2}{\sqrt{50}}}$$

$Z_{\frac{\alpha}{2}} \rightarrow Z$ score for $\frac{\alpha}{2}$ significance level

if

$|Z| > Z_{\frac{\alpha}{2}}$ then Accept H_1 or Reject H_0

- ethanol mixing changed the engine mileage significantly

$|Z| \leq Z_{\frac{\alpha}{2}}$ then Don't accept H_1 or Failed to reject H_0

- change in mileage due to ethanol mixing is not significant.

Test type	Z vs Z_α	Conclusion
Right tailed	$Z > Z_\alpha$	Accept H_1
left tailed	$Z < -Z_\alpha$	Accept H_1
two tailed	$\lvert Z \rvert > Z_{\frac{\alpha}{2}}$	Accept H_1